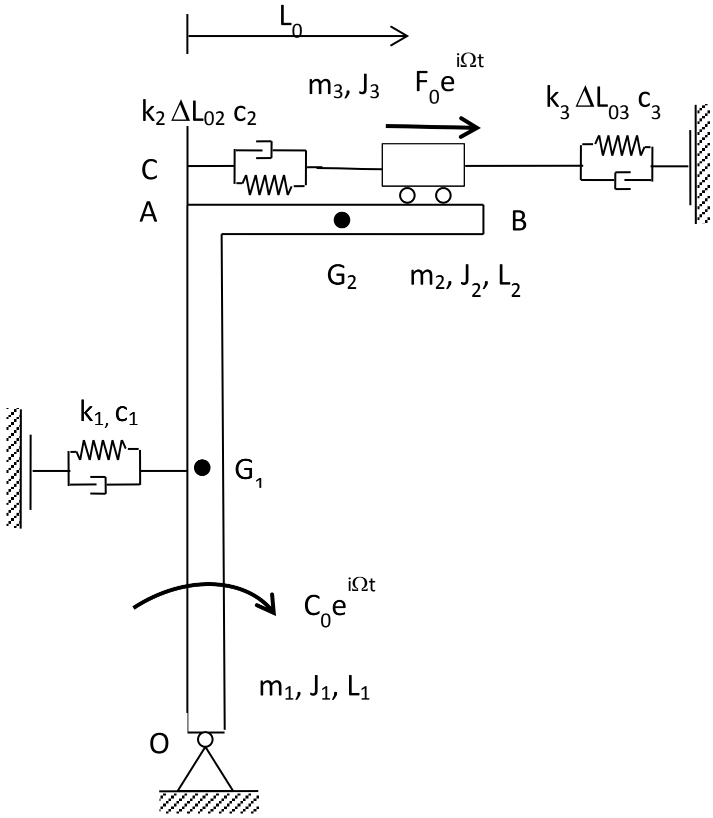
Mechanical Systems Dynamic

December 11th, 2014 – Version A



m_1 = 10 kg	k_1 = 3000 N/m	F_{01} = 100 N
m_2 = 5 kg	k_2 = 1000 N/m	C_{02} = 10 Nm
m_3 = 3 kg	k_3 = 1000 N/m	$\varphi_1 = \pi/6$ rad
J_1 = 0.2 kgm^2	ΔL_{02} = 0.02 m	$\varphi_2 = \pi/3$ rad
J_2 = 0.02 kgm^2	ΔL_{03} = 0.02 m	$\Omega_1 = 2\pi$ rad/s
J_3 = 0.01 kgm^2	c_1 = 10 Ns/m	$\Omega_2 = 4\pi$ rad/s
L_1 = 0.50 m	c_2 = 0.5 Ns/m	
L_2 = 0.25 m	c_3 = 0.5 Ns/m	
L_0 = 0.10 m		

The mechanical system above lays in the vertical plane and is shown in its equilibrium position. It is constituted by a "L shaped" rigid body and by a second rigid body (m_3, J_3), which is sliding on the horizontal side of the first body (consider negligible the distance AC). The rigid bodies are connected each other through a spring and damper (k_2 c_2). The static preload of the springs k_2 and k_3 is given and is equal to ΔL_{02} and ΔL_{03} respectively. In the static configuration the distance between the center of gravity of the mass m_3 and point C is L_0.

The students are requested to:

1. Write the linearized equations of motion around the given static configuration;
2. Compute the natural frequencies and modes of vibrations;
3. Compute the Frequency Response Function (in the range 0 – 5 Hz, step = 0.01 Hz) applying an external horizontal force at mass m_3. Output: the rotation of the "L shaped" rigid body and the force transmitted by the spring and damper system (k_3 c_3). Save the graphical results as SNxxx1.fig and SNxxx2.fig, whit S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your identification number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
4. Compute the static preload of the k_1 spring (write also the numerical result on the sheets of the solution);
5. Compute the time history of the vertical component of the constraint force of the hinge O, given the following inputs: a horizontal force $F_{01}e^{i(\Omega_1 t+\varphi_1)}$ applied to the mass m_3, and a torque $C_{02}e^{i(\Omega_2 t+\varphi_2)}$ applied to the "L shaped" rigid body. Save the results as SNxxx3.fig.

Note: Please collect the matlab script file relative to the numerical solution of the exercise and save it with the name SNxxx.m, where S states for the first letter of the surname, N for the first letter of the name and xxx are the last three digits of the identification number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (.fig e .m) have to be compressed (please select either the **.zip** or the **.rar** format) in a file named SNxxx.zip (o SNxxx.rar), which has to be uploaded to the beep portal of Politecnico di Milano. Each student is requested to:

1. Connect to the site https://beep.metid.polimi.it;
2. Enter the portal using its own identification code and password;
3. Select the section *Homework* of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select one of the following the folders *Test Dec. 11 2014 CS17- A*, *Test Dec. 11 2014 CS17- B*, *Test Dec. 11 2014 BL27- A*, *Test Dec. 11 2014 BL27- B,* according to the classroom and test version
5. Click on the button *Add* and select *Single document*;
6. Browse the compressed file SNxxx.zip using the button *Browse… (or Sfoglia…)*
7. Fill the *Title* field writing its own code SNxxx (e.g. for Bruni Stefano 123456: BS456);

Confirm the procedure clicking on the button *Publish*.